COLD FUSION TIMES

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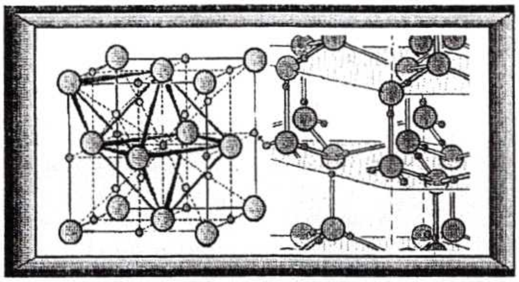
THE JOURNAL DEVOTED TO RELEVANT SCIENTIFIC AND ENGINEERING RESEARCH OF THE NEW HYDROGEN ENERGY



Metallic palladium (at left) is shown fully loaded with deuterium, an isotope of hydrogen. Heavy water (D_2O, on the right) is the source of the isotopic fuel used in some "cold fusion" reactions. Notice the hydrogen lattice within the palladium lattice.

ITALIAN COURT VINDICATES DRS. PONS & FLEISCHMANN

According to the Salt Lake Tribune, cold fusion pioneer, Dr. Stanley Pons, has just won his lawsuit that was filed against **Il Giorno**, a Milan (Italy) newspaper. The paper reported "two negative articles about the former University of Utah professor in November 1991". Dr. Pons' suit in the Civil Court of Milan was settled out of Court on May 11, 1994 "for a significant, undisclosed amount of cash for damages", The Salt Lake Tribune reported in its May 18, 1994 edition written by Paul Rolly and Joann Jacobsen-Wells. There remains yet another Italian court case pending involving the Italian newspaper, **La Repubblica** which incorrectly labelled cold fusion "scientific fraud". The plaintiffs include Drs. Fleischmann, Pons, Preparata, Bressani and Del Guidice. (Continued on page 5)

Events have been moving quickly in the field of cold fusion. **COLD FUSION TIMES** features articles with material data about cold fusion phenomena. More data on physics. More data on science, and on helium, and on electrode preparation, and even electrode dopants -- the field grows. The topic continues to expand.

This issue of the **COLD FUSION TIMES** continues to provide detailed information for our readers on the **products** of cold fusion and on **factors** influencing outcome of these reactions

FACTORS AFFECTING COLD FUSION

Dennis Cravens

[Ed. Dennis Cravens, the 1994 Reilly Citation Award winner (see COLD FUSION TIMES vol. 1 n. 3), presented his magnum opus at ICCF-4 and further discussed his observations in the recent premiere issue of COLD FUSION (see page 4). These important excerpts are from his work, and are critical for any scientist or scholar in this field.]

It appears that high loading ratios of deuterium to palladium are helpful, if not actually required, to achieve anomalous heating events. Although the dynamic conditions discussed below may cause temporary regions of high deuterium ratios in the metal, by and large the most important factor in achieving the heat is to achieve a large over-all deuterium/palladium (D/Pd) ratio. To achieve the large loading ratios, steps should be taken: (a) to select host lattice materials: (b) to prevent cracking of the host lattice; (c) to load the cathode in a uniform manner; and (d) to avoid contamination.

Selecting host lattice materials The quality of the metal host selected to contain the deuterium is important to achieving anomalous heat. Remember, the goal is to load deuterium (D) into the host cathode at a rate faster than the D can escape. This is equivalent to saying that before trying to blow up a rubber balloon one should make sure there are no holes in it. This means that conditions that increase the egress of the D should be limited and conditions that increase loading should be enhanced. (Continued on page 2)

HELIUM-4 (^4He) IS PRODUCED DURING COLD FUSION REACTIONS

Mitchell Swartz

The cold fusion phenomena are quite novel not only because they couple to nuclear reactions, but because they only occur in a physically dense material and not in a "gas-like" plasma such as the sun or a hot fusion reactor. As a result several new, fantastically different disciplines of material physics and engineering have developed in the solid state.

Why is this fusion different? These nuclear phenomena occur as special cases within material science, that of first-row interstitial lattices fully loaded to 1:1 (or greater) atomic ratios within another much heavier (e.g. palladium or certain other Group VIII or rare earth) lattice. It may even turn out that one key to the unusual reactions are the dissimilar nuclear weight lattices consisting of the heavy palladium and the very light deuterium. The other keys appear to involve matters of anharmonicity, periodicity, potential high loading of the tetrahedral sites (the so-called gamma phase), and electrode purity that make cold fusion quite complicated. **(Continued on page 7)**

"COLD FUSION"

Dr. Edmund Storms graces the cover of newly launched "COLD FUSION" magazine

"COLD FUSION" MAGAZINE LAUNCHED

New Hampshire - If imitation is the sincerest form of flattery, then the staffs of **COLD FUSION TIMES** and **FUSION FACTS** must smile when they see **"COLD FUSION"** magazine making its emergence. **"COLD FUSION"** is the offspring of Charles Becker, the financial backer of ENECO and Wayne Green of **Byte** and **73** magazine fame. We welcome **"COLD FUSION"** magazine to the forefront and look forward to hearing about more CF news. The magazine is glossier than most and effervesces with Editor Mallove's editorial touch. Although poorly received on the nets (page 10), we should pay attention to it as an **ABC-TV Good Morning America report** recently did when they gave it mention.

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TOWARDS THE FUTURE

PROBLEMS OF THE FARADAY MODEL

by Peter Graneau and Neal Graneau

[Ed. This all too brief excerpt is from "Newton versus Einstein - How matter Interacts with Matter", an excellent text by Drs. Peter Graneau and Neal Graneau]

The scientific principle which was to permit the large scale generation of electric power was discovered by Michael Faraday (1791-1867) in the laboratory of the Royal Institution in London. Some consider him the greatest of all experimental physicists. Quite apart from being a gifted experimenter, he took Oersted's electric conflict outside the wire seriously, and with it initiated a dramatic reversal of the philosophy of science from far-actions back to the contact actions of the middle ages. Son of a blacksmith, the twenty-one-year-old bookbinder Faraday had his appetite for science wetted by the books he helped to produce.

One day he approached the Director of the Royal Institution in London for a job. Sir Humphrey Davy, the noted physicist and chemist, made him his laboratory assistant. The Institution consisted of and still is today-a famous lecture theater and library devoted to the physical sciences, with laboratories in the basement. Fourteen years later, in 1827, Faraday became the Director of the Institution. He discovered electromagnetic induction in 1831. He found an electromotive force, or voltage, could be induced in a wire when a current-carrying circuit was moved past the wire, or when the current in a nearby stationary circuit changed with time. When a battery drove a current around a circuit it seemed natural that the driving voltage was propagated in the wire by contact action. But when the current was driven by an induced voltage, caused by a separate circuit, it looked rather like action at a distance.

The far-action treatment of electromagnetic induction was taken up by the German physicist F. E. Neumann. Before him, Faraday had formulated a very different theory of induction. Being a non-mathematical physicist, Faraday did not attempt to stipulate the law of electromagnetic induction in mathematical terms. He had his own intuitive way of reasoning. In his mind he saw a picture of imaginary lines of force linking the circuit, which caused the induction, to the circuit which experienced the induction. This was his way of establishing contact between the two circuits. It eliminated the need for far-actions and the mathematics which accompanied the action at a distance forces. Whenever induction occurred, the number of lines linking the two circuits was changing. (continued on page 4)

Because of rapidly developing technologies in cold fusion research, the COLD FUSION TIMES remains available for home or office delivery.

COLD FUSION TIMES brings up-to-date, relevant information to its readers, and an open forum for all views on cold fusion.

The COLD FUSION TIMES is not responsible for the comments of its reporters or readers whose words remain protected by the First Amendment of the U.S. Constitution.

Questions, comments, suggestions, and contributors are welcome.

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FACTORS AFFECTING COLD FUSION

continued from page 1

The proper loading techniques and the limitation of cracking are perhaps the most important factors in reproducing the anomalous heat. High purity (› 99.5%) Pd is recommended. I use Pd from Aldrich or investment-grade Pd from Englehart. It is especially important that it have low levels of platinum (Pt) impurities. The Pd should be visually inspected (or examined by sonic imagery) to assure that it is free from obvious cracks, deformations or Lutter lines. A quick dip in formic acid or an electrolyte (to wet the surface) often helps in visualizing the surface structure of the material.

I determined empirically that bubble patterns on the electrodes are often predictors of excess heat later in the experiment. A "good" piece of metal will initially take up deuterium very readily at low current density; few if any bubbles are seen on the surface. A "bad" sample often exhibits surface bubbles very early in the initial loading stages. It is sometimes disconcerting to see marked differences from samples cut from adjacent regions of the same wire or rod, but that is the nature of palladium.

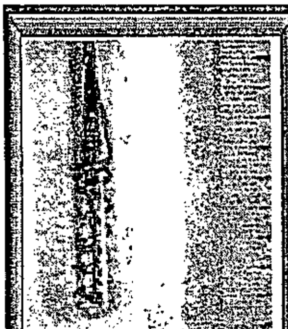
It is often possible to identify good regions of the Pd by its ability to form gas bubbles under water. I use clear quartz containers (B&L brand spectroscopic tubes) so that I can observe the cell's operation. If the current is momentarily stopped, the deuterium can be seen out-gassing. Uniform and very fine bubbles usually indicate a good piece of Pd. On the other hand, localized regions of large bubbles indicate regions where there are pathways for the D to rapidly deload the lattice. If you observe such regions, you should try again with another piece of Pd. Subsequent microscopic examination usually shows voids, cracks, or other deformations in such regions. You can eventually come to recognize potentially good pieces of Pd by observing bubble patterns. I developed a quantitative method of pretesting to catch the bubbles in inverted tubes. This allows me to determine the bubble volume as a function of unit cathode length. Also, I can determine which Pd samples are the best to use. The "best" samples will be the last to show bubbles at their surface perhaps the single best way to predict a sample's future success.

The time of onset of the bubbles is a better early predictor of a sample's performance. A sample which never bubbles until it is loaded to 0.6 to 0.8 D/Pd is a "very good sample." This indicates that the surface is not permitting recombination. If the cathode does not bubble early in the loading process. You are doing it right and you can use Faraday's Law to estimate loading. In an array of similar cathodes loading simultaneously the last one to bubble is the best.

It is also possible to screen samples by their volume expansion. Using a micrometer, a cathode sample is first measured. Then it is electrochemically loaded into its beta phase (0.6‹ D/Pd‹ 0.7). "Good" samples do not increase their volume more than 10%. "Bad" samples exhibit volume changes of 15% or more.

Preventing cracking of the host lattice Care must also be taken to prevent cracking of the metal host lattice after you have selected a good piece of Pd. A larger piece re quires greater care in loading and handling. The reason is that the Pd lattice expands as it loads. The loading is analogous to heating a piece of glass. Large pieces cannot withstand rapid expansions without cracking. However, smaller pieces can more easily withstand abrupt expansions. Thus, it is best to slowly load the Pd to avoid internal stresses which would result in increasing the D pathways out of the Pd. If cracks do develop, it is best to remelt and recast the Pd and start again.

It is very rare for Pd once it is cracked ever to be successful in demonstrating the anomalous heat. The cracking difficulty can also be overcome by use of alloys (such as 20% Ag in Pd, 5% Re or Rh in Pd, 5% V and Sn in Ti, etc). The 10-25% silver alloy of Pd resists cracking on deuterium absorption. However, it does lengthen the loading time due to decreasing the D diffusion rates.



Dr. Cravens'microelectrode

Adding Li can shorten loading times. Though difficult to form and costly, the 5% Rh in Pd alloy seems to be best suited for achieving anomalous excess heat. Another tactic is to plate Pd on Ag or another substrate such as Ni or Cu. The advantages are two-fold. First, the plating can be preformed to partially form Pd D. This avoids cracking that results from expansions during rapid loading. Second, it provides better thermal conductivity. (I recommend that the plating be done in an ammoniated solution and the film formed should be, at least 5 to 20 microns thick.) Pre-annealing (heating) and gas loading at elevated temperature seem to decrease the tendency of the lattice to crack. I often heat the Pd in a quartz tube. The tube and Pd is heated (in air) until just below the softening point (estimated 800-900°C). The idea is to reduce the internal stress of the Pd metal lattice by the annealing. After about two hours, the tube is brought to one atmosphere of pressure of D_2 gas to help partially load the Pd lattice. Continued on Page 5

PEOPLE IN THE NEWS

POWER PROFILE OF THE MONTH - Hal Fox

There are few people in the field of cold fusion who have contributed more to solidifying, validating, and encouraging research than Hal Fox. Aside from being the first editor of Fusion Facts, the first publication devoted to cold fusion, he has marathoned around the world touching base with scientists, businesspeople, engineers, academicians, and others in hopes of expanding research and helping negotiate the acquisition of patent applications. With his educational background in engineering, computer science, and business, he said he knew a good thing when he saw it.

"Because cold fusion was such an interesting area of science, it was exciting to know that these nuclear reactions can be controlled in such a modest environment," said Fox recently from his office at Fusion Information Center in Salt Lake City, Utah where he is President and CEO. His goal? Rounding up all the inventions in the field under one roof thereby smoothing the way for implementation and reducing the number of legal battles that may ensue. After five years of globetrotting from such destinations as Russia and Japan, Hal has continued to disseminate data in cold fusion to a point where every scientist and non-scientist alike can now read the facts, debate and learn from each other.

As co-chair of the International Symposium on Cold Fusion and Advanced Energy Sources Minsk, Belarus (co-sponsored by the Belarus State University) which took place May 24-26, he has brought together top researchers from around the world. Of the approximately 120 people who attended, there were about 50 presenters who moved this field ahead yet again. The "Cold Fusion Source Book" which is about to be published with Fox's guidance, will include selected papers in this field.

"The cleaness and availability of cold fusion is what intrigued me the most ... I was thoroughly schooled in the problem of our environment. I felt cold fusion was an answer to our environmental problems." Hal Fox

Armed with a B.S. in physics and math from the University of Utah in 1951, Hal Fox served in the Air Force from 1951-1959 during which time he studied meteorology (at UCLA). Later, he joined the Hughes Aircraft Company as a missile systems engineer ultimately directing University of Utah (at Research Park) research in fluid dynamics. But it wasn't until he earned his M.B.A. from Utah and later his Ph.D. in Computer Science that he became a computer "expert" and began a new company called C.A.I. (Computer Aided Instruction), a software group that primarily made computer programs for authors. He believes that programs can be designed for cold fusion as well. When the "announcement" was made in 1989, he said he couldn't sleep wondering about all the possibilities that could emerge especially in the computer realm. And he began accumulating increasing amounts of cold fusion research.

"If I was going to gather data I might as well publish it," said Fox. "That's when Fusion Facts began. Later I got involved with business development, got some people together, and founded Fusion Resources which later became F.E.A.T. (Future Energy Applied Technology). They eventually merged and ENECO was founded.

Now semi-retired from this last venture, Fox is Editor-in-Chief of Fusion Facts. His book "COLD FUSION IMPACT" is published in four languages.

- Gayle Verner

A challenge to the cold fusion skeptics:

by Hal Fox

If Drs Stanley Pons and Martin Fleischmann should be criticized, it would be for not stressing the difficulty of designing and operating a successful cold fusion electrochemical cell. Almost none of the criticism poured out on Pons and Fleischmann came from their peers, the world's foremost electrochemists. For example, Dr. John O 'M Bockris, a world renowned electrochemist, was one of the first to replicate the effect. Dr. Edmund Storms also achieved early successes. Operating on the assumption that if there were a nuclear reaction it had to be the fusion of deuterium-deuterium, the hot plasma physicists leaped to the conclusion that fusion in a metal lattice must produce the same results as d-d fusion in high temperature deuterium plasma.

Because most of the hot fusion physicists had immediate access to neutron measuring equipment, many of them hurriedly put together experiments, operated for a few hours, found no neutron emission, and then began labeling Pons and Fleischmann poor experimenters (at best) and frauds (at worst). Most of such early negative papers indicate that the authors, study of electrochemistry had been neglected.

Electrochemists, who served as consultants, had a depth of knowledge of this new procedure apparently gathered from little more than TV news footage. Obviously, what took Pons and Fleischmann several years of work was not replicated in a few days by those unskilled in the art. Frightened by the concept that two chemists could produce "excess power" in a relatively simple table-top experiment, some of the hot fusioneers (who have yet to achieve excess power with hot fusion devices) began an unprecedented, and vocal, attack on cold fusion in general, and Pons and Fleischmann in particular, an attack still waged by a few confirmed skeptics-two of whom attended the Fourth International Conference on Cold Fusion (Maui, Hawaii, December 1993).

The laughable part about this on-going skirmish is that the "pathological skeptics" are still denouncing the heavy-water., palladium, lithium system that Pons and Fleischmann announced March 23, 1 989. Many other scientists, recognizing a good thing when they learn about it, have forged ahead with their own ideas about other ways in which nuclear reactions can be produced and controlled in other types of devices.

Here is the latest scoreboard of devices that produce nuclear reactions (and nuclear byproducts including heat):

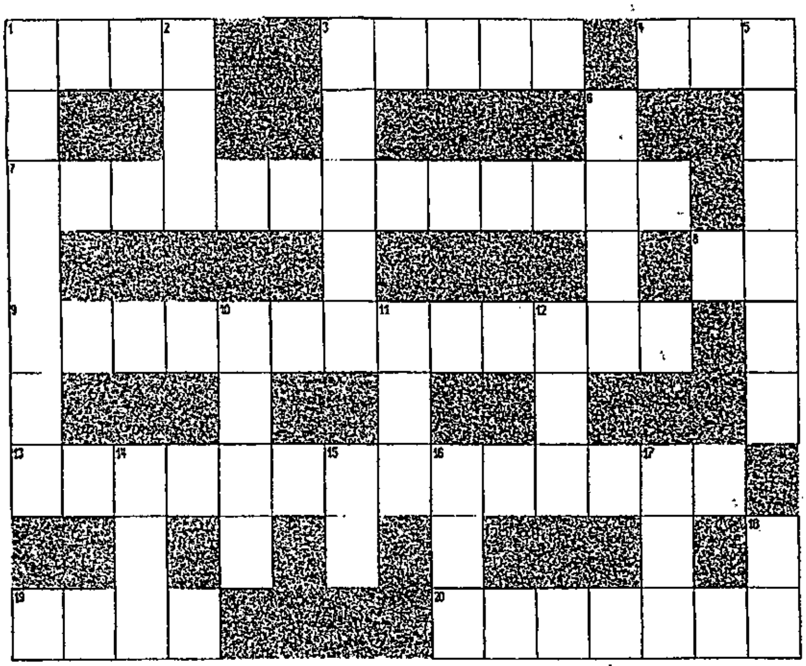
1. Molten salts using potassium, sodium, and lithium salts and a palladium electrode. (Liaw and Liebert, University of Hawaii, 1 990, Fusion Technology January 1993.)
2. Demineralized water (light water), nickel cathodes, and, for example, potassium carbonate electrolyte. (Randell Mills, 1989-90, Fusion Technology August 199 1, Bush and Eagleton, Cal Poly - Pomona, 1 99 1, Fusion Technology September 1 992.)
3. Glow Discharge (low pressure), palladium cathode, deuterium gas system. (Kucherov, Karabut, & Savvatimova, LUCH, Russia, 1990-94, Fusion Technology December 1992.)
4. Sparking discharges using deuterium gas and palladium cathodes. (Dufour, France, 1 989-90-93, Fusion Technology September 1 993.)
5. Capillary cold fusion. (Work funded and reported by J.P. Vigier, 1990-93, in former Czechoslovakia; Physics Letters A.)
6. Bronze crystals, sodium tungstate crystals which form tiny tubules. (Kaliev, Bamboshkin, Samgin, et al, 1992-93, Ekaterinburg, Russia, ICCF-3.)
7. Proton Conductors. (Mizuno, Enyo, Akimoto, & Azumi, Hokkaido Univ., Japan, 1993, ICCF-4.)

[Ed.: Hal Fox is a dean of this field in the United States, and we are pleased to run this excerpt from his article in COLD FUSION (v.1, n.1).]

COLD FUSION CROSSWORD #1 —

by Jorel Robertz

Answer next issue. The first reader to send (determined by date of receipt) the correct answer receives a **COLD FUSION TIMES** subscription.



ACROSS

1. Scientist vindicated in Italy from an unwarranted attack.
3. What the Zeeman effect does to optical spectroscopy lines.
4. The final impact of cold fusion upon its skeptics.
7. Moving deuteron quanta in palladium.
8. Not an AND.
9. Located between crystalline grains.
13. What one can keep a hot fusion plasma in.
19 An electrode changes this upon loading with deuterium.
20. He studied photon scattering.

DOWN

1. It can be in loaded into palladium.
2. Lab which continued cold fusion research (abbrev.).
3. A massive hot fusion laboratory.
5. A useful commodity.
6. A cold fusion company.
10. Energy from nickel electrodes systems may scale as this.
11. A previous US agency controlling energy.
12. Type of energy transfer from alpha particles (abbrev.) "high-___".
14. The source of deuterium in a Mizuno-type system.
15. A type of radiation emitted from active cold fusion electrodes.
16. The lattice structure of niobium.
17. All active electrodes may all be found to be in one of these.
18. First two letters of element "Sb".

Decision of the judges is final. Offer is void where such offers, crossword puzzles, or knowledge of cold fusion is prohibited or illegal.

COLD FUSION THEORIES *****

This issue continues this column on theories of cold fusion.

Edward H. Lewis is a Chicago, IL scientist with technical writing and superconductivity experience. He is developing theories involving anomalous physics phenomena and his hypothesis that anomalous phenomena are produced as sets at approximately 80 year intervals.

Part 2 PLASMOIDS AND COLD FUSION by Edward Lewis

It seems that much CF phenomena is plasmoid phenomena. Since it seems that substance is converting to plasmoids, perhaps one may identify substance as plasmoid phenomena also. If this is so, then substance may behave quite differently than people have thought. Plasmoids convert to light and electricity, and so it seems that light and electricity converts to plasmoids. Plasmoids clump and divide and dissipate, and so maybe substance does also, and so new elements and molecules are produced. In addition to substantial evidence that substance seems to behave this way, there is another reason to suspect that substance may be identified as plasmoid phenomena. It is that there is substantial evidence that galaxies are plasmoid phenomena, and if this is so then one would expect that the things in it that people have called substance and particles and atoms are plasmoid phenomena as well. These two reasons, if accurate, seem to be sufficient. Some evidence that substance and therefore cold fusion is plasmoid phenomena is that people have experienced plasmoid phenomena during electrolysis and discharge, and that the round holes and tunnels and grain shaped holes that people have found in their electrodes are evidence that substance converted to plasmoids and light and electricity, and that all or almost all the kinds of anomalous traces that Matsumoto has shown seem to be plasmoid phenomena.

Based on the phenomena that Matsumoto produced, the pictures of tiny, bright, and stationary BL and corona-like phenomena (see Fig. 9 of Ref. 7) he has shown, the visible BL-like phenomena (8) that exploded that he reports, the sparks that he observed (9) that left traces like those produced before during electrolysis and discharge, the pictures and descriptions of electrodes, and the traces, one may categorize CF phenomena as plasmoids. Important evidence is the round holes and the trails on and in emulsions (see Fig. 4a, trace "D" of Ref. 10 (I am not sure it is a hole, but I suspect it is a boring.) and Fig. 7a of Ref. 11) and electrodes (see Fig. 8 and Fig. 10 of Ref. 7) that Matsumoto produced by discharging and electrolysis, the round holes in electrodes that Liaw produced, the holes in electrodes that others produced, the empty areas in electrodes that are shaped liked grains that Matsumoto (12), Numata et al. (13) and Silver et al. (14) produced and the half-empty grains that Matsumoto produced, and the holes and tunnels and trails on and in electrodes that Silver et al. produced. These tunnels, round holes, and trail-like marks are similar to those that are produced by ball lightning phenomena, though ball lightning are associated with bigger effects. These tunnels, holes, and trail-like marks are also similar to those produced by the EV phenomena that K. Shoulders produced (see Ref. 6 of Part 1 of the article). Silver and his co-authors published their paper (14) in the December, 1993 issue of FUSION TECHNOLOGY, and they have reproduced the tunnels, round and grain shaped holes, and trail-like markings in metals that Matsumoto produced. These tunnels, holes, and trail-marks are evidence of the conversion and change of materials.

Matsumoto may also have shown a picture of a plasmoid or a plasmoid precursor; or maybe it was a plasmoid (see Fig. 8 of Ref. 11). There is something that seems to be round or spherical and in a round cavity of an electrode. It is about 180 micrometers in diameter.

Important evidence that both CF phenomena and substance in general are plasmoid phenomena is Yukimura and Matsumoto's experience of the production of electricity by apparatus (15). Also, Ohtsuki and Ofuruton, by using microwave radiation, produced a plasmoid (16) which continued to burn outside their apparatus even though the generator was turned off. They could not explain why it would do this. It had burned a hole though aluminum foil, and it seems to me that the plasmoid continued to exist because the aluminum had converted to be part of the plasmoid. I suspect that the round holes in electrodes and emulsions that Matsumoto produced and the round holes and tunnels (see Fig. 8 of Ref. 14) that Silver produced are due to the boring of BL-like phenomena, and that the grain-shaped holes (see Fig. 2a of Ref. 14) that they produced is evidence of the conversion of the grain to light or electricity or of the production of plasmoids.

Plasmoids may also travel through materials without boring and even without apparent effect sometimes, even if the plasmoids are very big. The plasmoids that Matsumoto has produced did this. Some of them traveled through the glass of the containers, the water, and the air, and this is major evidence to support my deductions. Matsumoto has also shown pictures of sectioned electrodes with what seem to me to be trail-like tracks (see the left side of the electrode shown in Fig 13a of Ref. 12), as if tiny BL-like phenomena traveled inside and left tracks. These marks do not seem to be due to the disappearance of the material of the electrode, but they may be due to a chemical change, like that which seems to have occurred around the empty grains that Numata et al. produced (13).

7) T. Matsumoto, "Experiments of One-Point Cold Fusion," FUSION TECHNOLOGY, 24, 332 (Nov. 1993). See also "Photo. 7: Traces of White Holes on Nuclear Emulsions," in I. Yukimura and T. Matsumoto, SEARCHING THE ENERGY OF THE UNIVERSE -- PHOTOGRAPHS OF AC ELECTROLYSIS OF WATER, Japan, March, 1993.

8) T. Matsumoto, "Cold Fusion Experiments by Using Electrical Discharge in Water," distributed at the ICCF4.

9) I. Yukimura and T. Matsumoto, "Observations of Luminescences, Sparks and Radiations During ac Electrolysis Cold Fusion," submitted to FUSION TECHNOLOGY (1992).

10) T. Matsumoto, "Searching for Tiny Black Holes During Cold Fusion," FUSION TECHNOLOGY, 22, 281 (Sept. 1992).

11) T. Matsumoto, "Observation of Meshlike Traces on Nuclear Emulsions During Cold Fusion," FUSION TECHNOLOGY, 23, 103 (Jan. 1993).

12) T. Matsumoto, "Microscopic Observation of Palladium Used for Cold Fusion," FUSION TECHNOLOGY, 19, 567 (May 1991).

13) H. Numata, R. Takagi, I. Ohno, K. Kawamura, and S. Haruyama, "Neutron Emission and Surface Observation During a Long-Term Evolution of Deuterium on Pd in 0.1M LiOD," THE SCIENCE OF COLD FUSION, Proceedings of the ICCF2, T. Bressani. E. Del Giudice and G. Preparata, eds., SIF, Bologna, Italy, 1991.

14) D. Silver, J. Dash, and P. Keefe, "Surface Topography of a Palladium Cathode After Electrolysis in Heavy Water," FUSION TECHNOLOGY, 24, 423 (December 1993).

-- Continued further --

(continued from page 2)

FARADAY'S PROBLEM - Graneau and Graneau

It seemed natural to Faraday to assume the induced voltage was determined by the change in the number of lines of force, per second, linking the two circuits. Since the lines of force could not be seen nor counted, Faraday's rule of electromagnetic induction remained a qualitative law. Even if we make up some rules for calculating the number of lines of force which link two circuits together, Faraday's model has a serious shortcoming.

When the second wire, in which the voltage is induced, is not a closed circuit, the number of lines which link the two conductors becomes indeterminate and makes the calculation of the induced voltage impossible. We know, and can demonstrate, that voltages are induced in open circuits. A *good* theory of induction should be able to specify them. A full quantitative explanation of electromagnetic induction had to wait till 1845, fourteen years after Faraday's discovery. This explanation was provided by Franz Ernst Neumann (1798-1895), a professor of physics at the University of Koenigsberg in East Prussia. He was the first of a breed of physicists who devoted most of their time to theory. Neumann did not adopt Faraday's lines of force and the associated contact action philosophy. He remained faithful to distant actions. His research on induction was the logical continuation of Ampere's investigation of electrodynamics. It was O'Rahilly who referred to the combined theory as the Ampere-Neumann electrodynamics. This theory was taught and used primarily in Germany and France. The key memoirs of the first electrodynamic theory were not translated into English. This is the reason why it remained unknown outside of continental Europe. The Ampere-Neumann electrodynamics had its heyday in the second half of the nineteenth century. Maxwell was a keen student of it and then made it obsolete.

It was during this period when the electric power industry was born. The major components of electric power systems-generators, motors, lighting equipment, transformers, and transmission lines were developed in conjunction with the old electrodynamics. It would be wrong to suggest that the theory paced the evolution of electrotechnology. It was rather the other way around. The theoreticians tried hard to keep up with trial-and-error advances and inventions. Neumann's law of electromagnetic induction dealt with forces on the electric fluid in metallic conductors. It differed from the ponderomotive force laws. Yet there were also common aspects.

For example, Neumann's law applied to two matter entities, and specifically to two wire elements. In the last resort, as in the case of Ampere's law, the wire elements had to be conductor atoms, but the atomic theory of matter was still to come. Neumann's law of induction was also a simultaneous far-action law. However the electromotive force was not applied to the matter element, of the interacting pair of elements, but to the electric fluid inside it. The ponderomotive force laws described two-way actions. Both elements of matter were the cause of an effect in the other. Neumann's law of induction was a one-way law. One matter element, which had to carry electric current, was the cause of the electromotive force induced in the other element. The second element did not have to carry current, and therefore it could not react back on the first element. The electric fluid was not matter and forces on it, measured in volts, did not have to obey Newton's third law of equal and opposite reaction forces.

Another important difference between the law of induction and the ponderomotive force laws was that induction always involved "time," whereas the ponderomotive force laws did not contain the quantity known as time. For example, the induced electromotive force was proportional to the relative velocity between two conductor elements, and velocity stands for the distance covered per unit time. The induced voltage was also proportional to the time rate of change of the inducing current. Perhaps Neumann's most striking discovery was that the strength of induction does not fall off with the inverse square of the distance between the two conductor elements. *The induced voltage decreased in proportion to the inverse* distance. By doubling the distance between the interacting elements, the Ampere interaction is reduced to one-quarter, while Neumann's induction is reduced to one-half. The practical consequence of this is that induction reaches much further than the Ampere force.

COLD FUSION TIMES

COLD FUSION -- A Few Comments
Dr. E. Mallove, Sc.D., Editor "COLD FUSION"

[Dr. Eugene Mallove, cold fusion historian and scientist, has recently made some thoughtful recent comments ["COLD FUSION", May 1994 issue, vol 1 number 1 issue.]

"Cold fusion has now reached a critical stage in which improved communications will play a key role. The field is in ferment and expanding explosively. In one of history's classic ironies, the 1989 announcement of "excess energy from water" in a relatively simple table-top experiment-possibly by a heretofore unknown form of nuclear energy-occurred less than 12 hours before the Exxon Valdez caused a massive oil spill into the waters off the coast of Alaska.

There was an initial media hoopla over the cold fusion story, but the press then lost interest as it became more difficult to discern the truth amid claims and counter-claims of angry chemists and physicists. With few exceptions, journalists bought the notion that cold fusion was nothing but hot air. "Pathological science" became the common insult, as few noticed that pathological skepticism about a new phenomenon was the real problem. Contrary to the media's perception, cold fusion never died and was certainly never disproved; it simply went underground as groups of courageous scientists in over a dozen countries mounted a concerted effort to understand and reproduce the mysterious phenomenon. Thanks to their hard work, it has survived."

FACTORS AFFECTING COLD FUSION

Cravens (continued from page 2)

As a sample is annealed, impurities such as Pt, Cd, etc. can migrate to the surface. So if you anneal, be sure to grind and polish the surface to remove impurities after the annealing. Otherwise, the sample will not properly take up deuterium. This is because Pt, for example, catalyzes hydrogen recombination with oxygen. The deuterium will simply recombine at the surface and bubbles will be observed early in the loading process. Annealing is not required, but if you do it you must always follow it with polishing and surface preparation.

<u>Loading the cathode uniformly</u> The shape and surface texture of the cathode is also important. The cathode should be shaped to avoid sharp or jagged corners. Such points do not allow proper loading to be achieved. I find slightly better success with cylindrical shapes with rounded bottoms or narrow plates with rounded corners. The idea is to remember to minimize electrical-field (E-field) gradients across the surface of the host lattice. Such variations could lead to unequal loading. The D would tend to diffuse through the Pd and then escape from the lattice from areas with the least surface E-field directed to accelerate the positive deuterium ions into the lattice. Rough surfaces also tend to limit the loading. In my view, this is one of the greatest sources of confusion in understanding variations in success with different samples. Deep grooves or valleys normally are the regions with the least surface electrical potential (the smallest absolute value of surface potential). I polish the Pd to avoid such difficulties. This is done by first polishing on a buffer wheel. The purpose is to round any burrs, rough areas or sharp corners. The Pd is then finely polished with a series of finer aluminum oxide powders and finally with cerium oxide (Aldrich, optical grade) on a cotton cloth. Rough surfaces on the anode should also be avoided. Burrs and points seem to be sites of oxidation and corrosion (this has been seen with nickel anodes). I sometimes use an entire "spiral loop" as an anode with both ends outside of the cell. If you have a free end, you may wish to place it in glue (silicon rubber, epoxy, etc.) to prevent corrosion from the point when the cell is run for extended times. The best results seem to come with Pt anodes, but if you use Ni it should be oxidized first (by flame heating or with hydrogen peroxide solution).

It is important that the initial loading of the Pd be done slowly and carefully. The object is to use a low current density (with respect to the cathode surface areas, 30-60 mA/cm^2) so that there will not be unequal expansion and the development of large internal stresses. Do not be tempted to raise the current above 100 mA/cm^2 until the D/Pd ratio within the lattice is at least above 06-0.7. If you use electrical pulsing techniques, do not raise the peak current levels above the 100-200 ma/cm2 region until the Pd is loaded to at least D/Pd @ 0.65. You can calculate the amp-seconds needed to load the lattice to 0.7 or better. I prefer to load un til much longer, or about 150 amp-hours have passed for each cubic centimeter of Pd (with at least one dimension of the Pd less than 1 mm). Be patient! It is better to spend too much time in the initial loading stage than too little. Any sudden application of large currents before the cathode reaches the beta phase is likely to crack the host lattice. This could lead to rendering the Pd useless until it is recast. In some early work, I was unable to observe anomalous heat because of rapid initial loading and then using the cracked and stressed Pd over and over again.

A series of experiments were conducted to see the effect of. various initial loading rates on the percent excess power produced (all compared when running at 50°C and 500 mA/cm2).The cathodes that were slowly loaded out-performed those that were initially loaded at a high current density. It should be noted that the rapidly loaded wires had a greater volume expansion.

<u>Additives</u> It would seem that one role of the additives that have often been used in cold fusion experiments is to increase the so-called over-potential and the internal fugacity within the lattice and to select certain electrochemical pathways. Do not add "poisons" or other additives (Al, thiourea, Si, B, Mo) until you are loaded above 0.6-0.7 D/Pd. This could cause a creeping or spreading of any pre-existing cracks or voids.

The expansion of the lattice on absorbing the D may help to seal such voids. Thus, the goal is to first load slowly to allow the lattice to "heal" any deformations. Materials which can increase the internall fugacity should be added after most deformations are self corrected. A really good host lattice will not swell more than about 10% excess volume on loading. The shape of the anode can also affect the results. The anode should have the proper geometry to load the metal host uniformly. In a spiral-wound Pt anode, use enough turns to supply a uniform E-field at the cathode. This is similar to trying to increase the internal pressure of a balloon; a few points of pressure do not suffice. It takes uniform pressure from all sides.

Avoiding some kinds of contamination Impurities can prevent excess heat generation. I clean the anode and the electrolyte by running it in a separate system with a dummy electrode. This removes any reactive species and most impurities from the platinum and plates them onto a dummy cathode. It also removes some impurities that may be in the electrolyte. The dummy cathode is then removed and with it any such impurities. If Ni is used as an anode, it should first be oxidized to render its surface passive.

Continued on page 8

THE PATENT PAGE -- Analysis of the Intellectual Property Field in Cold Fusion

Normally, NEWS from the INTERNET is only on the penultimate page of the COLD FUSION TIMES. However, for this issue, a comment moving across the transom is appropriate. Paul Koloc wrote the following :

" .. *(Ludwig Plutonium) writes: › " ... Because now the PAF has all intellectual property rights to SPONTANEOUS NEUTRON MATERIALIZATION DEVICES plus in an indirect way as the outline of the physics of the art of SNMD, all rights to the correct theory of superconductivity. Thanks due to Mr. Behrend and the USA patent office that LP and PAF will not be "nailed to the cross of just 17 piddle paddle years".*

"Harvey Behrend may have had his name officially used to make patents relating to nuclear energy appear as ordinary patents, but, indeed, unlike other patents, there is a bit more afoot then this poor well utilized examiner will usually admit to. It is the DoE that actually determines the technical merit of nuclear related patents and not the patent office, although the patent office can still impose the rules of form and procedure. Those queries from Behrend are just a pass through from the DoE examiners, who are quite ignorant of the legal basis of patent law, unfortunately. Patents filed this year would normal issue in a year or two, but those that make DoE nervous can take as long as eighteen years! All the better for the inventor. Such practices tend to get beaten back by the Court of Appeals, as time goes on. For example, to prevent any and all fusion patents from issuing, they claimed that since no fusion reactor existed no one could be sure what they were, so they couldn't be patented!!! What dum-dums! The C of A had a gut busting laugh over that argument."

[Paul M. Koloc (pmk@prometheus.UUCP); Subject: Re: Patent 07/737,170, Patent Office, re: Harvey E. Behrend Message-ID: ‹Cps1KG.6Cv@prometheus.UUCP› ; 14 May 1994]

COLD FUSION HEATS UP - Storms
(continued from page 9)

Experiments by Yan Kucherov and co-workers at Russia's Scientific Industrial Association (LUCH) in Podolsk produced an electrical discharge in low-pressure deuterium, using palladium as one of the electrodes. Heat was accompanied by an unusual form of gamma radiation. Gamma rays ordinarily fan out in all directions from their source. In this case, the radiation emerged from the palladium in tightly focused beams. Such beams suggest the existence within the metal of reflection planes with much closer spacing than normally exist in the atomic lattice. Such planes might be owing to an unusual, tightly bound electron structure. Much of the evidence for cold fusion has come from techniques alien to nuclear physics, such as electrolysis and the precise measurement of heat. But in the past several years, researchers have been producing cold fusion results using the tools of nuclear physics, such as high energy ion beams.

Toshiyuki Iida and co-workers at Osaka University, for example, bombarded palladium and titanium samples with energetic deuterium nuclei. During this bombardment, the cell emitted a variety of ions with energies consistent with conventional fusion between two deuterium nuclei. This was in itself unremarkable, and could be explained as "hot" fusion, as high-energy deuterons in the beam slammed into those that had already taken up residence in the metal. But defying expectations, emission of energetic ions continued from the palladium target for many hours after the implantation beam was turned off. The Osaka group also detected high-energy helium nuclei. Such observations suggest that fusion-cold fusion-continues in the material even without the presence of a high-energy beam.

A trail of evidence for cold fusion comes from work by Jirohta Kasagi and co-workers at Tohoku University in Japan. Kasagi bombarded titanium deuteride with 150-kiloelectron-volt deuterons. When the deuterium content of the titanium deuteride was high enough, the experiment produced protons with energies up to 17.5-MeV These energetic protons are thought to result from the fusion of deuterium and helium-3. It is the helium-3 that provides evidence for cold fusion.

In theory, helium-3 could arise in two ways. One is from the " hot" fusion of the energetic deuterium nuclei with the deuterium contained in the metal. But the helium-3 is found in isolated pockets within the titanium, which is inconsistent with this explanation. If the helium were formed in hot fusion, it is Unlikely to diffuse through the material, as is required for the build up in pockets. More likely is that the helium-3 is a product of the radioactive decay of tritium, which was produced within the metal in a cold fusion process. The tritium thus formed could, unlike helium, easily migrate.

COLD FUSION TIMES

Thoughts about ICCF-4 and COLD FUSION
by Dr. Robert A. Huggins

"Except for a few ill-considered, and rather personal critical remarks by one of the attendees, the conference's tenor was quite normal for an area of newly-emerging science that has possible technological overtones. Earlier results in a number of areas were reinforced by more thorough and quantitative experimental work, and the critical parameters that are responsible for several of the earlier problems have now been established. The spectrum of experimental methods and observations has also been extended considerably. Nevertheless, there are still a number of major unresolved questions relating to the reasons for some of the important experimental observations.

The theoretical basis underlying the now well-established experimental documentation that nuclear processes can take place in solids under conditions that are far removed from those typical of hot plasmas is still far from mature. One of the complicating features is that the reactions expected from hot plasma experience are not found, and that different processes evidently dominate"

[brief excerpt of Dr. Huggins' ICCF-4 review in "COLD FUSION", May 1994 issue, vol 1 number 1]

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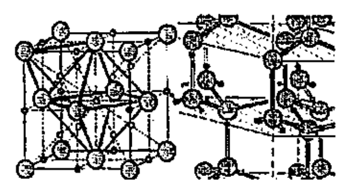
Our Favorite Cartoon of the Month

by Mike Smith
[LAS VEGAS SUN

"Perspectives"
NEWSWEEK



COLD FUSION TIMES

MATERIALS CORNER -- This issue: Helium, Sulfur, Yttrium

YTTRIUM, NIOBIUM and SULFUR

The Chinese (Jin) and the Japanese (Mizuno (December 1993) have recently published on the use of niobium, yttrium, and other materials for the cold fusion reactions.

"The cold fusion phenomena were tested with use of proton conductor solid electrolyte plates maintained at 300~400 deg C. An anomalous level of excess heat evolution of the order of 100 watt cm^{-2} was observed during absorption/desorption cycles of deuterium-containing hydrogen gas under application of an alternate electric field. **** Samples were made from a mixture of $SrCO_3$ CeO_2'Y_2O_3 and $0Nb_2O_3$ powders. **** The heat generation from the proton conductor in the experiment of deuterium-containing hydrogen gas was estimated to be approximately 50 watt (~ 100 watt cm^{-2}) over 20 hrs, or ~ 3.6 MJ in total. The input power given to the sample was + 18 V, +/- 40 (micro) A, or 7.2 x 10^{-4} watt. Accordingly, the output-to-input power ratio was estimated to be as large as 7 x 10^4. ["Anomalous Heat Evolution from SrCeO3-Type Proton Conductors during Absorption/Desorption of Deuterium in Alternate Electric Field"; Tadahiko Mizuno, Michio Enyo, Tadashi Akimoto and Kazuhisa Azumi Hokkaido Univ., Sapporo, Japan (ICCF-4, December 1993)]

"The experimental studies of YBCO-D system indicated that YBCO high temperature super-conductor (HTSC) was shown to have a similar effect on deuterium absorbability and anomalous nuclear effect like palladium(1). We found that $Y_1Ba_2Cu_3O_7$-d could absorb deuterium at normal temperature and forms $D_xY_1Ba_2Cu_3O_7$-d. We also found that the deuterated YBCO could produce high energy charged particles far larger than background. The influence of the absorbed deuterium on the characteristic of YBCO HTSC and the mechanism of the anomalous nuclear effect are not clear and needed to be further studied." ["DEUTERIUM ABSORBABILITY AND ANOMALOUS NUCLEAR EFFECT OF YBCO HIGH TEMPERATURE SUPER-CONDUCTOR"; JIN Shang-xian, ZHAN Fu-xiang and LIU Yu-zhen, en Beijing, PRC (ICCF-4, December 1993)]

Helium-4 from COLD FUSION

Continued from page 1

One important result of the coupled light-weight hydrogen lattice within the heavy palladium lattice is that the phonon (quantum of sound energy) spectra actually become split into acoustic and optical components. This separation may be the physical event which allows the coupling of the nuclear reactions to the lattice, with satisfaction of the conservation of momentum which is required.

As a result of this uniqueness, there have now been scores of experimental confirmations of new reactions producing both nuclear products and thermal energies (called excess enthalpies) created by these cold fusion phenomena. We begin this series with a discussion of one of the major products (or nuclear "ashes") of this new technology - helium-4 (^4He). This column (continuing next month on helium contamination issues) will briefly review some of the remarkable findings which clearly demonstrate that the helium-4 "ash" production by cold fusion is consistent with some of the reactions having a probable nuclear origin.

Back to the product - helium-4 What is most compelling is that the helium-4 production is both time-correlated with the onset of excess heat, correlated to the observed power levels of excess heat, and even correlated with the exposure of undeveloped x-ray films which are blanketed around some active electrodes. It is these multiple factors -- obtained in several labs -- which are strong in support of this finding of helium generation. These factors are thus also consistent with a nuclear origin to some of the cold fusion phenomena. (continued next issue)

The use of sulfur materials near/on the electrodes has been discussed by IMRA (Japan) [Akita (ICCF-4, December 1993), Hasegawa (December 1993), Hayakawa (December 1993), and Tsuchida (December 1993)] using THU a sulfur containing material. THU is thiourea [$S=C-(NH_2)_2$] where the sulfur is postulated to form the site of closest-approach to the palladium surface. (More on this next issue)

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FACTORS AFFEECTING SUCCESS

Cravens - (continued from page 5)

On the other hand, the connections to the cathode should be well below the surface of the electrolyte. Remember that the highest loading ratio achieved by your cathode is limited, in part, on the portion of the surface of the cathode which has the least inward tendency to have a lower threshold than do the marginal cathodes. It would seem that the good cathodes leak less (perhaps due to fewer cracks for deuterium to leak out) and can reach higher ratios at lower current densities.

It also seems that the initial slow loading is best at lower temperatures and low current densities. However, the heat production should be done at an elev ated termperature and at higher current densities.

One can make the analogy to filling a bucket having a hole. If the bucket has only a small hole, it can be filled with a small inflow. If the bucket has a large hole or several small holes, it can only be filled with a very large flow. Using higher current densities works, in part, by increasing the inflow above the outflow of the deuterium. Holding the current low during the initial stages works by decreasing the number of "holes" from which the deuterium will leak when one goes to the high current densities after the initial loading.

A few rapid pulses of current may also set up localized regions of high deuterium ratios within the cathode. The use of dynamic cur rents that are more rapid than the deuterium diffusion times can lead to high D/Pd ratios in some regions of the cathode.

Pulsing the electrolytic current

Changing the electrolysis current can set up dynamic variations and nonequilibrium ratios of the deuterium within the cathode. One way to do this is to pulse the electrolytic current. The goal is to set up a condition that would cause a crowding of the deuterium within regions of the cathode. To do this, I simply short out some of. my current to ground to cause a pulse.

It seems best that the pulsing is done while keeping a DC bias on the current. In other words, the pulses should be added on top of a DC bias. The bias is selected (I use 60 mA/cm^2) so that the cathode is not in a reverse-bias condition, but is always the cathode during the pulse. This prevents any de-loading that could occur if. it was rendered anodic. For those new to cold fusion work, it is recommended that they first slowly load (30 mA/cm^2 surface area for 150 amp-hours per cubic centimeter of Pd), then slowly ramp the current to 60 mA/cm2 (over two hours), and finally start pulsing between the 60 mA/cm^2 and 500 mA/cm^2 levels. The pulses should have a fast rise time and a period between a few seconds to a few hours long. I use a period of 30 minutes and a 50% duty cycle.

It is also possible to pulse the current through 0 volts and then reverse the current direction. If this protocol is used, it is important that the Pd never be deloaded past the beta phase and into the alpha phase. (The reverse voltage level should be kept low (< 0.15V) so that oxides do not form on the surface.) This method may be somewhat better than the above-described pulsing method. However, the DC bias method is recommended until experience is gained in estimating loading ratios from amount of current passed through the cathode.

FOR THE WORLD (more information)

COLD FUSION — Storms

(Continued from Page 10)

surface impurities is also important to produce the required deuterium content. McKubre's first experiments, which were successful, used pyrex containers for the cell. When he changed to Teflon, the cell did not work. Further investigation revealed the apparent reason: aluminum and silicon atoms from the pyrex migrated to the palladium surface and allowed the deuterium to build up to higher concentrations.

McKubre's studies also identified another criterion for producing excess heat. His work, which have now been supported by at least four other laboratories, indicates that the electrical current entering the cell must exceed a threshold of at least 150 milliamperes per square centimeter of palladium. For experiments using large pieces of palladium, this can mean a very large overall current. Many experimenters have failed to produce anomalous effects because they have not applied enough current to their cells. Several experiments have shown that the nature and timing of the electrical input to a cell significantly affects the chances of its success. Heat is more likely to be produced when the electrical current is repeatedly " ramped", from a high to lower level or briefly pulsed to high values. Applying certain microwave frequencies is also beneficial. Moreover, for reasons still unknown, electrolysis must continue for several days before the cell "turns on" and begins to generate excess heat. Many of the early efforts to reproduce the effect failed at least partly because the researchers did not stick with them long enough.

Experiments that use crackfree palladium and follow the proper procedures now routinely result in excess heat, nuclear products, or both. Heat might be arising from nuclear reactions other than fusion, as well. At least 1 0 different laboratories have measured significant excess energy coming from light-water cells using nickel, rather than palladium, as the cathode, and in which is dissolved a carbonate of one of the so-called alkali metals (lithium, sodium, potassium, or rubidium). Experiments by Robert Bush and Robert Eagleton at California Polytechnical University and by Reiko Notoya at Hokkaido University in Japan suggest that this heat arises from a transmutation reaction: a proton enters the nucleus of the dissolved metal to give the next higher element in the periodic table. Potassium, for example, is proposed to take up a proton to produce calcium, and rubidium is similarly transformed into strontium. Researchers have detected calcium and strontium at levels that suggest this transmutation is occurring.

Accumulating Evidence: Helium - One of the products of the fusion of two deuterium atoms is helium, which with its two protons is the next element up on the periodic table. Many experiments have produced measurable helium. Hawaii's Liaw for example, found significant helium in the palladium in his heat-producing experiments with a molten-salt electrolyte. Workers at Texas A&M also detected normal helium-4 in the palladium after tritium was produced. However, no effort was made to measure heat during this study. Q.F. Zng and colleagues at the University of Science and Technology in Chengdu, China, have detected helium-4 in titanium rods that produced excess heat in an electrolytic cell. They saw no helium when excess heat was absent, suggesting a link between the two.

Skeptics have complained that cold-fusion experiments have produced too little helium; if the heat measured truly has nuclear origin, they contend, far more helium should be detected. The helium found in the palladium after the Hawaii experiments, for example, while significant, could account for only 10 percent of the large amounts of excess energy that this work generated. Recent work has whittled away at this objection. Melvin Miles and Benjamin Bush of the Naval Air Warfare Center at China Lake, Calif., for example, have measured helium in the gas emitted by electrolytic cells and have measured helium levels 90 percent of those needed to account for the measured heat. Since these scientists found no helium when using normal water in the cell instead of deuterium-bearing heavy water, the heat appears to be nuclear in origin. When critics suggested that the helium entered the glass collection flasks from the atmosphere, Miles and Bush repeated the experiment using steel flasks and obtained similar results while also showing that the diffusion rate of helium through glass was too small to account for the observed amounts. Apparently, most of the generated helium resides in the gas, rather than in the electrodes where researchers have mainly been looking.

When helium forms during fusion, two particles are condensed into one. The laws of physics require that in any interaction between particles, momentum (the product of mass and velocity) must be conserved. In hot fusion, a 24-MeV gamma ray is emitted to satisfy this conservation requirement. No gamma rays with this energy are observed in cold fusion. The most probable reason, I believe, is that the material has transformed into a special condition that enables the atomic lattice to absorb most of the nuclear energy being generated. If we assume that the material in this condition has a similar ability to absorb nuclear radiation applied from outside, a straightforward experiment could test this hypothesis. One could irradiate the material with gamma radiation, which easily penetrates ordinary palladium. When this special condition exists, the material should block some of the gamma rays; the portion of gamma radiation absorbed should correspond to the amount of the material that has switched to the fusion-enabling condition. Evidence for this special condition does exist.

(Continued on page 6)

LETTER CORNER

[including News from the INTERNET (*)] -- see also page 6

The following are two excerpts from the Internet regarding the Technology Review article (Storms, page 10) and "COLD FUSION" magazine. This is presented for balance as a parallax view. The first comment is from Mark Hittinger (bugs@NETSYS.COM) 5 May 1994

"Hi out there - just wanted to make a few comments about the recent article in "Technology Review" (edited at MIT!). One of the local alumni hereabouts cornered me and showed me the article. He was quite shook up about it. For those who haven't read it yet it is by Dr. Edmund (we are all gonna be *RICH*) Storms. It is a review of the past couple of yearsof positive results. Anyhow this alumni type was bugging me with a bunch of questions such as "is anything new going on?" "have there been any new breakthroughs that nobody wants to talk about?" "come on you went to Hawaii - just how many palladium call options do you have???" "just exactly where are P&F now??" Of course I replied that I thought it was almost dead but not quite dead yet. I told him about heat after death and ruined the rest of his day."

"This fellow loved to joke about CNF from time to time and the article appearing in his school's magazine messed him up. He is/was afraid that he missed something. Frowns now instead of jokes. I have to tip my hat to Dr. Storms for making an excellent public relations move. The article was probably too long. It should have been shorter so that it could be xeroxed and faxed around easier. I have seen complaints from some readers about the technical content of the article - but my point is that the technical content is a minor issue compared to the fact that the article appeared in the *MIT* alumni magazine. (Dr. Jones take note!)"

"Maybe Wayne can send Eugene and Jed to PR school now that they are going to rake in big bugs off all those reader service cards. If we can't have straight shooting science at least we can have warm fuzzy PR instead of conspiracy theories."

After this, the following reply was posted [6 May 1994]

"Mark I liked your rebuttal of the "Cold Fusion"/Mag thing, but you weren't vehement enough :)

I found my copy at the U of Wash bookstore (a fine bookstore) and got hours of amusement. The cost ($10) and the ads for Palladium medallions (for investment purposes) tell me where the money is to be made in CF." - RM Holt

ITALIAN COURT VINDICATES DR PONS

Continued on page 9

Dr. Douglas O. Morrison, hot fusion spokesman for CERN and point-man against cold fusion, is a self-purported expert-witness for the defense in the second case. He has published over the Internet that "La Repubblica, which is sometimes regarded as the New York Times of Italy, wrote an article in October 1991, reviewing the book by Axel Kuhn, "False Prophets" where it was suggested that Cold Fusion was an example and was "scientific fraud".

Profs. Fleischmann, Pons, Bressani and Del Guidice have sued La Repubblica for 8 billion lire which was about $5 million. Morrison reports that the Judge appointed as technical adviser, Prof. Licheri of Cagliari. Briefs were submitted by Prof. Gozzi of Rome for the Plaintiffs and from Morrison of CERN for the Defendants. Astonishingly, other professional skeptics have gone so far as to advertise their fax numbers on the Internet to offer their support for the Defense (for example, Dr. Frank Close of the U.K.)

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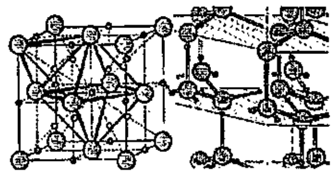
COLD FUSION TIMES

TO COLDLY GO WHERE NO ONE HAS GONE BEFORE



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COLD FUSION HEATS UP

Edmund Storms
MIT TECHNOLOGY REVIEW
[Brief excerpt from May 1994 issue]

The demand for long-term control cells reflects results from the early cold fusion experiments, which produced heat or nuclear products in occasional bursts. More recent work has achieved steady output. Once a cell turns on, it typically stays on for many days. Skeptics who accept the data on excess heat nevertheless assert that it originates not from any nuclear process but from some hitherto unknown chemical reaction. But no evidence from any study has been reported to support various speculated chemical sources. If a novel chemical reaction is the source of this power, significant quantities of some chemical product must be present in the cell, yet no such products have been observed. Moreover, many distinct and novel chemical reactions would be required to explain the excess-heat observations in a the variety of chemical environments that have now been studied.

One reason that cold fusion has been so difficult to accept is that the experiments are so hard to replicate. Many experienced researchers, attempting to follow exactly the methods described in published studies, have, until recently, produced little or no evidence for a nuclear reaction. But scientists are starting to solve the puzzle of why some cells work and some do not.

An important requirement for producing heat, for example, now appears to be palladium largely free of microscopic cracks. Deuterium apparently escapes from any cracks too fast for a critical concentration to build up. Michael McKubre and co-workers at SRI International, as well as researchers at several other laboratories, have shown that the larger the ratio of deuterium to palladium in the electrode, the greater the heat. The presence of

Continued Page 11

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